inflation —*a smooth, uniform spatial expansion populated by a nearly uniform distribution of matter*—was exactly what we were trying to explain. It's exactly the low-entropy configuration that we need to explain time's arrow.

Entropy and Inflation

Indeed, this is significant progress. But two important issues remain.

First, we seem to be concluding that the inflationary burst smooths things out and hence lowers total entropy, embodying a physical mechanism—not just a statistical fluke—that appears to violate the second law of thermodynamics. Were that the case, either our understanding of the second law or our current reasoning would have to be in error. In actuality, though, we don't have to face either of these options, because total entropy does not go down as a result of inflation. What really happens during the inflationary burst is that the total entropy goes up, but it goes up *much less than it might have.* You see, by the end of the inflationary phase, space was stretched smooth and so the gravitational contribution to entropy—the entropy associated with the possible bumpy, nonordered, nonuniform shape of space—was minimal. However, when the inflaton field slid down its bowl and relinquished its pent-up energy, it is estimated to have produced about 10^{80} particles of matter and radiation. Such a huge number of particles, like a book with a huge number of pages, embodies a huge amount of entropy. Thus, even though the gravitational entropy went down, the increase in entropy from the production of all these particles more than compensated. The total entropy increased, just as we expect from the second law.

But, and this is the important point, the inflationary burst, by smoothing out space and ensuring a homogeneous, uniform, low-entropy gravitational field, created a huge *gap* between what the entropy contribution from gravity was and what it might have been. Overall entropy increased during inflation, but by a paltry amount compared with how much it *could* have increased. It's in this sense that inflation generated a low-entropy universe: by the end of inflation, entropy had increased, but by nowhere near the factor by which the spatial expanse had increased. If entropy is likened to property taxes, it would be as if